

CS329H: Machine Learning from Human Preferences

Stanford Frontier AI Learning System

Baseline: October 6, 2026. Print companion to the HTML course site:
course map, prerequisites, notation and shapes, glossary,
cheatsheet, crash course.

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Course map

Baseline 2026-10-06. Units U01-U05 built by the first builder. Units U06-U10 plus consolidation built by the second builder 2026-10-07.

Official schedule (verified 2026-10-07)

Session	Date (2026)	Title	Unit map	Claim class
S01	Wed 23 Sep	Course introduction and motivation	U01 background	Source-supported
S02	Mon 28 Sep	Choice data, generalization, and Bradley-Terry	U01-C01-C09	Source-supported
S03	Wed 30 Sep	Random utility models and axiomatizations	U02-C01-C06	Source-supported
S04	Mon 5 Oct	Factor models	U02-C07-C09	Source-supported
S05	Wed 7 Oct	Learning to predict choice. Latent-variable estimation with MLE	U03-C01-C03	Planned
S06	Mon 12 Oct	Bayesian estimation and finetuning	U03-C04-C07	Planned
S07	Wed 14 Oct	Fisher information and optimal experimental design	U04 (all)	Planned
S08	Mon 19 Oct	Reinforcement learning from human feedback	U05-C01-C06	Planned
S09	Wed 21 Oct	Direct preference optimization	U05-C07-C08	Planned
S10	Mon 26 Oct	Active elicitation for LLMs, and case studies	U06	Planned
S11	Wed 28 Oct	Assistance games. Assisted vs. contextual bandits	U06	Planned
S12	Mon 2 Nov	Acquisition functions and Thompson sampling	U07	Planned
S13	Wed 4 Nov	Dueling bandits and preferential Bayesian optimization	U07	Planned

Official unit headers on the page: Foundation (S01-S04), Learning (S05-S09), Action (S10-S13), Inversion (S14-S15), Aggregation (S16-S18).

Concept-to-session provenance in this build

- U01: C01-C09 map to S01/S02 titles (source-supported). C10 annotation protocol, C11 social meaning, C12 fairness questions are requested extensions. Taught as independent theory, labeled PLANNED / SOURCE ATTRIBUTION PENDING.
- U02: C01-C06 map to S03 (source-supported). C07-C09 map to S04 (source-supported). C10 scale ambiguity, C11 estimation assumptions, C12 counterexamples are requested extensions. Independent theory, labeled pending.
- U03: S05/S06 are planned. All twelve concepts taught as independent theory, labeled pending.
- U04: S07 planned. All twelve concepts independent theory, labeled pending.
- U05: S08/S09 planned. All twelve concepts independent theory, labeled pending.
- U06: S10/S11 planned. All twelve concepts independent theory, labeled pending.
- U07: S12/S13 planned. All twelve concepts independent theory, labeled pending.
- U08: S14/S15 planned. All twelve concepts independent theory, labeled pending.
- U09: S16/S17 planned. All twelve concepts independent theory, labeled pending.
- U10: S18 planned. All twelve concepts independent theory, labeled pending.

Assessment on the official page (learning objectives, not reproduced work)

Project 50%: pre-analysis plan (4 Nov) then final manuscript plus code (3 Dec). Three required disclosures (integrity, reflection, impact). Quizzes 30%: three 15-minute closed-book quizzes. Orals 20%: two 12-minute oral exams. This course builds original equivalent practice only. It never solves or reproduces assessed work.

Prerequisites

Shared bridges live at `workspace/stanford-frontier-ai/v2-pack/shared/prerequisites/` and are linked, not rebuilt. Each unit below carries its own local remediation.

Prerequisite graph for owned units

Unit	Bridges	Why
U01	P06, P07, P10	Bernoulli outcomes, likelihood, train/test split
U02	P03, P06, P07	Embeddings, Gumbel/softmax, axioms as constraints
U03	P07, P09, P18	MLE, gradient descent, priors/posteriors, EM
U04	P05, P07, P18, P22	Score as gradient, curvature as Hessian, design as experiment
U05	P08, P17, P18	KL, policy gradients, reward as latent variable

Local remediation per unit (self-contained, inside the lessons)

- U01 lesson opens with a Bernoulli refresher: events, PMF, expectation, variance, IID. No step in U01 uses a symbol before the refresher defines it.
- U02 lesson opens with a dot-product and norm refresher plus a Gumbel CDF derivation from scratch.
- U03 lesson opens with the Bayes rule and log-likelihood refresher and derives the EM lower bound inline.
- U04 lesson opens with gradient/Hessian refresher and derives the score identity inline.
- U05 lesson opens with a KL and policy-gradient refresher and derives the DPO reparameterization inline.

Diagnostic (closed book, 20 minutes)

Answer in prerequisites.md only after attempting. Keys at the end of this file.

1. A coin lands heads with probability 0.6. Write its PMF, expectation, and variance.
2. State Bayes rule in words and in symbols.
3. Define the log-likelihood of n IID Bernoulli trials.
4. What does it mean for a parameter to be non-identifiable? Give a one-line example.
5. Compute the gradient of $f(x) = x^2 + 3x$ with respect to x .
6. Define KL divergence between two discrete distributions in one line.
7. What is the difference between a training objective and an evaluation metric?
8. Write the shape of a matrix that maps a 4-vector to a 3-vector.

Diagnostic keys

1. PMF: $P(X=1)=0.6$, $P(X=0)=0.4$. $E[X]=0.6$. $\text{Var}=0.6*0.4=0.24$.
2. Posterior equals likelihood times prior divided by evidence: $P(A|B) = P(B|A)P(A)/P(B)$.
3. $l(p) = \sum_i [y_i \log p + (1-y_i) \log(1-p)]$.
4. Two different parameter values give the same data distribution, so data cannot tell them apart. Example: Bradley-Terry scores shifted by a constant.
5. $2x + 3$.
6. $KL(p||q) = \sum_x p(x) \log(p(x)/q(x))$.
7. The objective is what the optimizer minimizes. The metric is what the stakeholder uses to judge success. They can disagree.
8. 3 by 4.

Not-yet-understood dependency list (course level)

1. Measure-theoretic probability, not needed. Finite discrete cases suffice here.
2. Variational inference beyond the ELBO statement, introduced locally in U03.
3. PPO internals, compared at the objective level in U05, not re-derived.

Notation and shapes

One symbol table for U01-U05. A symbol keeps one meaning across units. First-use page in parentheses.

Choice data

Symbol	Meaning	Shape / units	First use
D	preference dataset	set of n records	U01
(a, b, y)	one comparison: item a, item b, outcome y	a, b in [m]. Y in {0,1}	U01
y = 1	a chosen over b	dimensionless	U01
n	number of comparisons	positive integer	U01
m	number of items	positive integer	U01

Scores and utilities

Symbol	Meaning	Shape / units	First use
s_i, theta_i	latent score of item i	real scalar, utils	U01
U_i	random utility of item i	real scalar	U02
u_i	deterministic part of utility	real scalar	U02
epsilon_i	random shock	real scalar, same units as u	U02
Delta	score gap s_a - s_b	real scalar	U01

Models

Symbol	Meaning	Shape / units	First use
$\sigma(z)$	logistic function $1/(1+\exp(-z))$	scalar in (0,1)	U01
$P(a > b)$	probability a beats b	scalar in [0,1]	U01
$L(\theta)$	likelihood	positive scalar	U01
$l(\theta)$	log-likelihood	real scalar	U01
π	policy, distribution over outputs	function or vector	U05
π_{ref}	reference policy	same shape as π	U05
$r(x, y)$	reward of response y to prompt x	real scalar	U05
β	KL penalty strength	positive scalar	U05

Statistics

Symbol	Meaning	Shape / units	First use
$s(\theta)$	score function, gradient of log-likelihood	vector, $\dim(\theta)$	U04
$I(\theta)$	Fisher information matrix	$\dim(\theta)$ by $\dim(\theta)$	U04
Var_{hat}	asymptotic variance estimate	scalar or matrix	U04
$p(\theta D)$	posterior	density over θ	U03
$p(y_{\text{new}} D)$	posterior predictive	scalar in [0,1] for binary y	U03

Embeddings

Symbol	Meaning	Shape / units	First use
w_j	respondent j embedding	real vector in $\mathbb{R}^d$	U02
v_i	item i embedding	real vector in $\mathbb{R}^d$	U02
d	embedding dimension	positive integer	U02

Conventions

- Vectors are column vectors. Probabilities are dimensionless and live in $[0,1]$.
- Log means natural log unless stated.
- "utils" marks an arbitrary latent scale. Only gaps and ratios of gaps are meaningful.
- Seed, dtype, and shape are stated for every computed example: float64, fixed seed 0 unless noted.

Glossary

Term	Definition
Preference	A latent ordering over items: what the person would want if fully informed. Not directly observed.
Choice	The observed pick from a set of options in a context. The data we fit.
Pairwise data	Records of the form (a, b, y): item a faced item b, outcome y says which was chosen.
Ordinal	Information about order only: A beats B. No statement about how much.
Cardinal	Information about strength: A beats B by a stated amount.
Bradley-Terry	Model: $P(a \text{ beats } b) = \sigma(s_a - s_b)$. Scores s are latent, estimated from pairwise outcomes.
Logistic likelihood	Product over comparisons of $\sigma(y * \text{gap})$ terms. Its log is the training objective.
Identifiability	Whether data can pin down a unique parameter value. Bradley-Terry scores are identified only up to an additive constant.
Random utility	$U_i = u_i + \epsilon_i$: deterministic value plus a random shock. The choice is argmax of U .
Gumbel	Noise distribution whose argmax over in-

Value of information	Expected gain in decision quality from an answer before asking: $E[\max_a E[U \mid \text{answer}]] - \max_a E[U]$.
Assistance game	The agent is unsure of the human's objective and learns it from the human's actions. Shared payoff, asymmetric knowledge.
Contextual bandit	The agent sees a context, then picks an arm. The best arm depends on the context.
Assisted bandit	The agent proposes. The human can veto or override. The team payoff exceeds either alone.
Thompson sampling	Sample a world from the posterior. Act optimally for the sample. $P(\text{pull } a) = P(a \text{ optimal})$.
Acquisition function	A score over candidate queries or arms (expected improvement, information gain) that ranks what to try next.
Dueling bandit	A bandit that learns from pairwise comparisons (duels) instead of absolute rewards.
Strong regret	Per-round shortfall of the duel's best Copeland score versus the maximum Copeland score.
Copeland score	An arm's average pairwise win probability against the other arms.
Preferential Bayesian optimization	Optimization using only pairwise comparisons, no absolute evaluations.
Reward inversion	Inferring the latent reward from observed choices under a rationality model.
Bounded rationality	Humans choose with noise, bias, and computational limits. The choice rule is not exact maximization.
Misspecified rationality	The assumed choice model (temperature, bias form) is wrong, so

Cheatsheet

Formulas, definitions, and numbers for all ten units. Provenance: S01-S04 source-supported (title level), S05-S18 PLANNED / SOURCE ATTRIBUTION PENDING.

Formulas

Name	Formula
Bradley-Terry	$P(i \text{ beats } j) = 1 / (1 + \exp(-(s_i - s_j)))$
BT log-likelihood	sum log sigmoid($y * \text{gap}$) over pairs
BT identifiability	scores identified up to +c, anchor (sum zero or first zero)
Softmax (Gumbel-max)	$P(i) = \exp(u_i) / \sum \exp(u_j)$
IIA	$P(A)/P(B)$ independent of C
KL-constrained optimum	$\pi^*(y x)$ proportional to $\exp(r(x,y)/\beta) \pi_{\text{ref}}(y x)$
DPO loss	$-\log \text{sigmoid}(\beta * (\log(\pi(yw)/\pi_{\text{ref}}(yw)) - \log(\pi(y_l)/\pi_{\text{ref}}(y_l))))$
DPO implicit reward	$\beta * \log(\pi(y x)/\pi_{\text{ref}}(y x))$
Value of information	$E[\max_a E[U \text{answer}]] - \max_a E[U]$
Bayes update (tea toy)	$0.8 * 0.5 / (0.8 * 0.5 + 0.3 * 0.5) = 0.727$
Strong regret	$C^* - \max(C_i, C_j)$, C = true Copeland scores
Jury theorem	$\sum_{k > n/2} C(n,k) p^k (1-p)^{n-k}$, needs $p > 0.5$ and independence
CI (95%)	mean $\pm 1.96 * \text{SD} / \sqrt{n}$
Borda (3 candidates)	2/1/0 points per ballot
Pareto frontier (toy)	$(\sqrt{t}, \sqrt{1-t})$, t in [0,1]

Key numbers

Where	Number
U01 BT gap, 5-1 toy	1.61
U03 EM, Gaussian mixture	-775.9 to -738.2
U06 Vol toy	0.25
U06 greedy vs eps-greedy regret	29.6 vs 2.4 (T=300, seed 0)
U07 TS vs greedy vs eps-greedy	4.0 vs 89.6 vs 5.2 (seed 23)
U08 inversion interval	[-0.94, 3.31], sign unsure
U09 cycle margins	+1/+1/+1
U09 cycle frequency	0.081 (11 voters, seed 0)
U09 spoiler	A 4-3-2, then B (A2 D2 B3 C2), D loses 2-7 to B
U09 jury	0.6826 (n=5), 0.9791 (n=101) at p=0.6
U09 subgroup	pooled 0.73, groups 0.87 / 0.59
U09 welfare t=0.5	(0.707, 0.707)
U10 capstone A	uniform 17.24 [16.56, 17.92], TS 3.11 [2.23, 3.99], leader 8.36 [6.86, 9.87]
U10 capstone B (HYPOTHETICAL)	funnel 6000/6000/5664/5664, drift 0.793 to 0.718, BLOCK

Definitions

- **Identifiability:** distinct parameters give distinct data distributions.
- **IIA (Arrow):** the A-vs-B social order depends only on A-vs-B ballots.
- **Strategyproofness:** no voter gains by misreporting.
- **Strong regret:** per-round shortfall of the duel's best Copeland score vs the max.
- **Pre-analysis plan:** hypotheses, metric, sample, success rule, failure rule, frozen before the run.
- **Contribution:** baseline, intervention, evidence, scope.
- **Negative-result report:** hypothesis, numbers, diagnosis, what it rules out.
- **Data consent:** purpose, voluntariness, withdrawal, data-use limits, retention.
- **Oral ladder:** define, toy, derive, implement/complexity, compare, debug, critique, design.

Axiom and escape maps

- **Arrow:** UD, Pareto, IIA, non-dictatorship. Escapes: 2 candidates, restricted domain (single-peaked), cardinal ballots, randomization.
- **Gibbard-Satterthwaite:** 3+ outcomes, deterministic, non-dictatorial rules are manipulable. Escape: random dictator (strategyproof in expectation).
- **Jury:** needs independence and $p > 0.5$. Correlated errors void it.
- **Interpersonal comparison:** needs a common scale. Rescaling one voter flips sums.

Honest-number rules

1. Every computed number states its seed.
2. CIs are procedure properties, not bounds on the truth.
3. Hypothetical numbers stay labeled HYPOTHETICAL.
4. Negative results are reported with numbers, not buried.
5. The constant in BT scores is meaningless, anchor before quoting.

Crash course

A fast pass over all ten units. One page per unit: the core ideas, the key numbers, the failure modes, the interview line. Provenance: S01-S04 source-supported (title level), S05-S18 PLANNED / SOURCE ATTRIBUTION PENDING, taught as independent theory.

U01. Preference models

Core: choices reveal preferences, the Bradley-Terry model turns pairwise wins into scores via the logistic function. Scores are identified only up to an additive constant: anchor before reporting.

Numbers: 5-1 toy gives MLE gap $\log(5) = 1.61$. $P(A \text{ beats } B) = 1/(1+\exp(-\text{gap}))$.

Failure: noise attenuates the gap, unanchored scores drift across runs.

Interview line: "BT scores are differences, the constant is meaningless."

U02. Choice theory

Core: random utility models (Gumbel noise gives softmax). IIA: the A-vs-B ratio does not depend on C. Axioms (transitivity, IIA) are testable restrictions.

Numbers: IIA predicts red-bus/blue-bus shares $0.50/0.25/0.25$, violations break it.

Failure: cycles in pairwise data cannot be fit by any score vector, the residual is the signal.

Interview line: "IIA is a prediction, not a law."

U03. Estimation

Core: MLE for choice models, EM for latent variables, Bayesian posteriors with priors, calibration of predicted probabilities.

Numbers: Bernoulli mixtures are flat for EM (verified -277.0138 everywhere), the Gaussian mixture climbs -775.9 to -738.2.

Failure: flat likelihoods (non-identifiability) stall optimizers, the Bernoulli flatness is the documented case.

Interview line: "Check the likelihood surface before trusting the optimizer."

U04. Policy optimization

Core: KL-constrained optimum π^* proportional to $\pi_{\text{ref}} * \exp(r/\beta)$. Fisher information measures what the data can identify. Design criteria (A/D-optimality) rank experiments.

Numbers: the closed form is exact under the KL constraint, off that assumption it is an approximation.

Failure: reward hacking (the proxy rises, the truth falls), differentiation under the integral needs regularity.

Interview line: "The optimum is closed-form only under the KL assumption."

U05. RLHF and DPO

Core: RLHF trains a reward model then optimizes against it (PPO). DPO skips the reward model: the implicit reward is $\beta * \log(\pi/\pi_{\text{ref}})$, trained with a logistic loss on pairs.

Numbers: DPO costs about 2x SFT per pair, no RL loop.

Failure: reward hacking, off-policy data breaks the DPO derivation, a bad reference policy drags the optimum down.

Interview line: "DPO has an implicit reward, it is not reward-free."

U06. Active elicitation and assistance games

Core: value of information (expected gain in decision quality). Assistance games: the agent is unsure of the human's objective and learns it from actions. Contextual vs assisted bandits.

Numbers: Vol toy = 0.25 (posterior best 0.75 minus prior best 0.5). Tea/coffee posterior 0.727. Greedy regret 29.6 vs epsilon-greedy 2.4 (T=300, seed 0).

Failure: greedy locks onto the wrong arm, elicitation burden caps real queries.

Interview line: "Information is worth what it changes."

U07. Acquisition, Thompson sampling, dueling bandits

Core: Thompson sampling (sample from the posterior, act optimally for the sample). Dueling bandits learn from pairwise comparisons, strong regret measures the duel quality. Acquisition functions rank what to try next.

Numbers: 4-arm BT bandit, strengths 1.2/0.8/0.3/-0.5, seed 23: greedy regret 89.6, epsilon-greedy 5.2, TS 4.0. Capstone: uniform 17.24, dueling TS 3.11, leader TS 8.36 (T=200, 20 seeds).

Failure: random duels never trigger the 0.95 stopping rule in 500 rounds (leader-focused duels trigger at round 43).

Interview line: "Sample the world, then act optimally in the sample."

U08. Inversion, style, bounded rationality

Core: invert observed choices to latent rewards under a rationality model. Style vs content: the reward mixes both. Bounded rationality: humans choose with noise and bias.

Numbers: inversion interval $[-0.94, 3.31]$: the sign is unsure, and the lesson says so.

Failure: misspecified rationality (wrong temperature, wrong bias model) inverts to the wrong reward, the empty interval means the model contradicts the data.

Interview line: "The interval is the honest answer, the point estimate hides the uncertainty."

U09. Aggregation, impossibility, fairness

Core: Arrow (no ranked rule satisfies UD, Pareto, IIA, non-dictatorship with 3+ candidates). Gibbard-Satterthwaite (every such rule is manipulable). Condorcet cycles, spoilers, interpersonal comparison, jury theorem, subgroup fairness.

Numbers: cycle margins $+1/+1/+1$, cycle frequency 0.081 (11 voters, seed 0). Spoiler: A wins 4-3-2, D enters (loses 2-7 to B), B wins. Jury: 5 voters at 0.6 = 0.6826, 101 voters = 0.9791. Subgroup: pooled 0.73 hides $0.87/0.59$. Welfare: $t=0.5$ gives (0.707, 0.707).

Failure: correlated voters void the jury theorem, unnormalized scales flip welfare sums.

Interview line: "Pick your poison deliberately, document the dropped axiom."

U10. Research projects and oral mastery

Core: pre-analysis plans, literature review, falsifiable hypotheses, data consent, method baselines, uncertainty quantification, reproducible code, contribution statements, integrity/reflection/impact, oral formats, negative results, open-source gaps.

Numbers: capstone A (H1 supported, H2 not), capstone B (hypothetical audit: BLOCK on drift).

Failure: burying the negative result, quoting hypothetical numbers as real.

Interview line: "The failure is data, report it with numbers."

The arc in one paragraph

Model preferences (U01-U02), estimate them (U03), optimize policies (U04-U05), elicit actively (U06-U07), invert carefully (U08), aggregate honestly (U09), research reproducibly (U10). Every claim has a home unit, every number has a seed, every failure is reported.